VEDULA ABHISHEK

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LinkedIn | GitHub | Portfolio

Research & Professional Experience

Research Fellow — CVIT, IIIT Hyderabad March 2026 – Present

- Applied YOLO to vehicle detection and drone video analysis within a Drone Analytics Management System.
- Worked with VLMs and LLMs on zero-shot evaluation, parameter-efficient fine-tuning (PEFT), custom adapters, and visual question answering (VQA).
- Investigated road infrastructure deficiencies using visual events, IMU motion-sensor data, GPS, and Indian Roads Congress (IRC) rules, aiming to link braking or swerving to road conditions, location, and applicable rules.
- Worked on temporal change detection by comparing videos of the same road stretch captured at different times.
- Developed a custom vision-language model (VLM).

Data Science Intern — AI Variant Dec 2023 – Jun 2024

- Prepared and analyzed datasets using Python and Pandas, performing data cleaning and exploratory data analysis to support machine learning experiments.
- Built Streamlit applications for model predictions and visualizations; collaborated on model tuning and communicated findings through reports and presentations.

Projects

Drone-Based Road Asset & Vehicle Detection GitHub
Python, PyTorch, YOLO, TensorFlow Lite, VLMs, Ollama

- Trained and benchmarked YOLOv8, YOLO11, and YOLO26 on an HPC cluster using JATAYU (10 classes) and DRASHTI-HaOBB (14 classes), covering horizontal and oriented bounding-box detection.
- Exported YOLOv8n and YOLO11s to INT8 TensorFlow Lite; built a web application with model comparison, FPS monitoring, VLM scene queries, and CSV reporting.

Adversarial Robustness of CNNs GitHub
Python, PyTorch, torchvision, torchattacks, CIFAR-10

- Conducted a controlled study of Gaussian smoothing, residual denoising, and pooling modifications to adversarially trained ResNet-18 models using a shared training recipe.
- Evaluated PGD-20 robustness on 10,000 test images at a pixel-space perturbation budget of 8/255 and APGD-CE/APGD-DLR robustness on a 1,000-image subset.
- Reported 49.18% PGD-20 accuracy for the plain ResNet-18 baseline versus 46.61% with all three modifications; documented evaluation differences between normalized and pixel-space attack budgets.

Brain Tumor Classification GitHub
Python, TensorFlow/Keras, OpenCV, Streamlit

- Built a CNN for tumor/no-tumor MRI classification using image preprocessing and augmentation; developed a Streamlit interface for interactive inference.

Technical Skills

Languages: Python, SQL
Multimodal AI & NLP: VLMs, LLMs, PEFT, Custom Adapters, Zero-shot Evaluation, VQA
Retrieval: Retrieval-Augmented Generation (RAG), Embeddings
Machine Learning & Vision: YOLO, CNNs, Object Detection, Adversarial Training, OpenCV, Model Evaluation
Frameworks & Libraries: PyTorch, TensorFlow/Keras, scikit-learn, Hugging Face, Pandas, NumPy
Tools: GitHub, Jupyter, Google Colab, Streamlit, Docker, Matplotlib, Seaborn

Education

M.Tech, Computer Science and Engineering (AI) 2024 – 2026
Centre for Development of Advanced Computing (C-DAC), Mohali CGPA: 9.0

B.Tech, Computer Science and Engineering 2019 – 2023
Bhilai Institute of Technology, Raipur CGPA: 8.8

Achievements

Paper accepted at ICVGIP 2026